EduTrack — Documentation

1. Project Description

EduTrack is a learning project for managing students and courses. The system is implemented as a REST API (backend) and a single-page web application (frontend). This document describes the architecture, installation, API endpoints, and working with the frontend.

Backend (REST API)
Frontend (single-page web application)

2. Technologies
FastAPI
PostgreSQL
SQLAlchemy
Pydantic
HTML, CSS, JavaScript

3. Project Structure
main.py — entry point
database.py — DB connection
models.py — models
schemas.py — validation
routers/ — API
frontend.html — UI

4. Backend запуск
Run the server:

```
uvicorn main:app --reload
```

Open Swagger:

http://127.0.0.1:8000/docs

5. API
Students
GET /students
POST /students
PUT /students/{id}
DELETE /students/{id}
Courses
GET /courses
POST /courses

PUT /courses/{id}
DELETE /courses/{id}
Enrollments
GET /enrollments
POST /enrollments
DELETE /enrollments/{id}

6. Data Models
Student: id, name, email, age
Course: id, title, credits
Enrollment: student_id, course_id

7. Frontend
Open:

edutrack-frontend.html

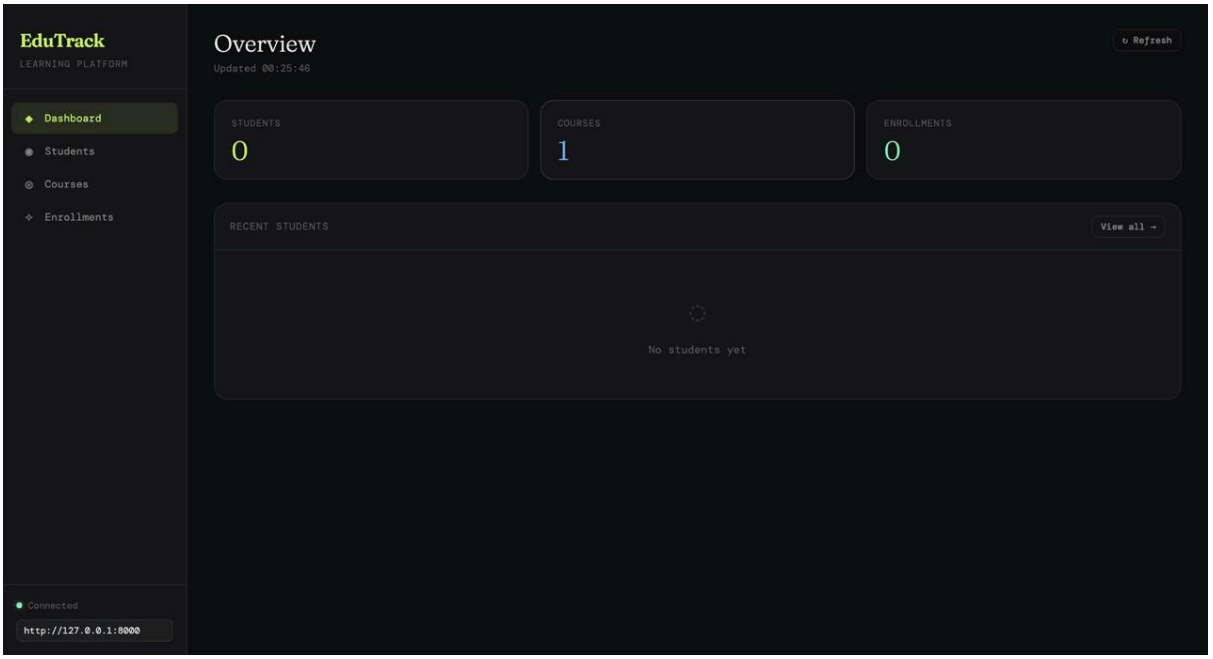
Set API URL:

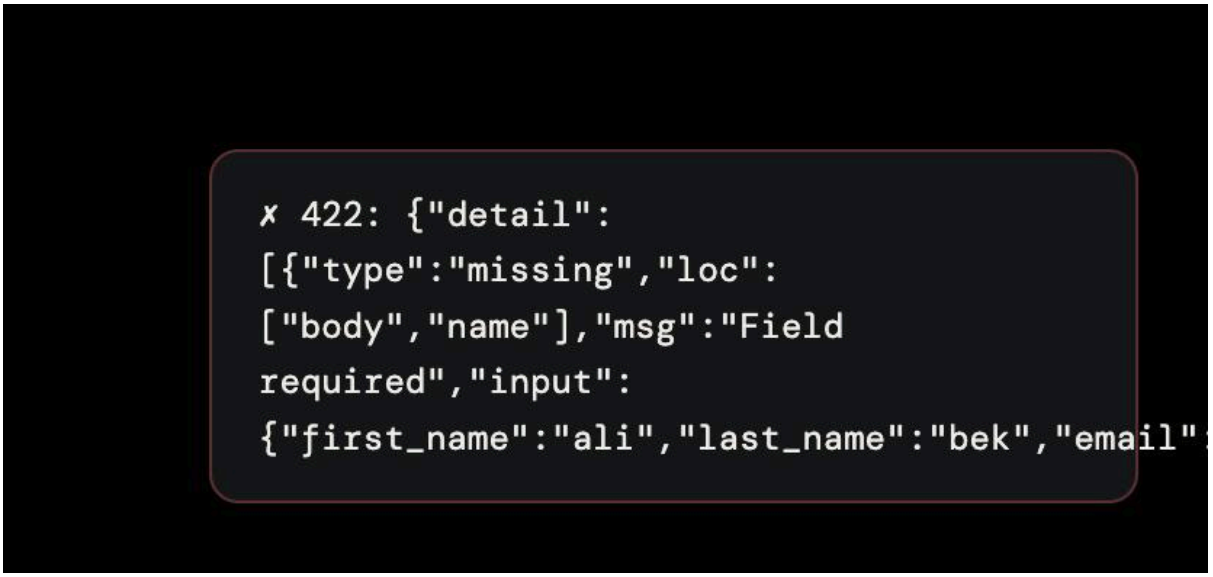
http://127.0.0.1:8000

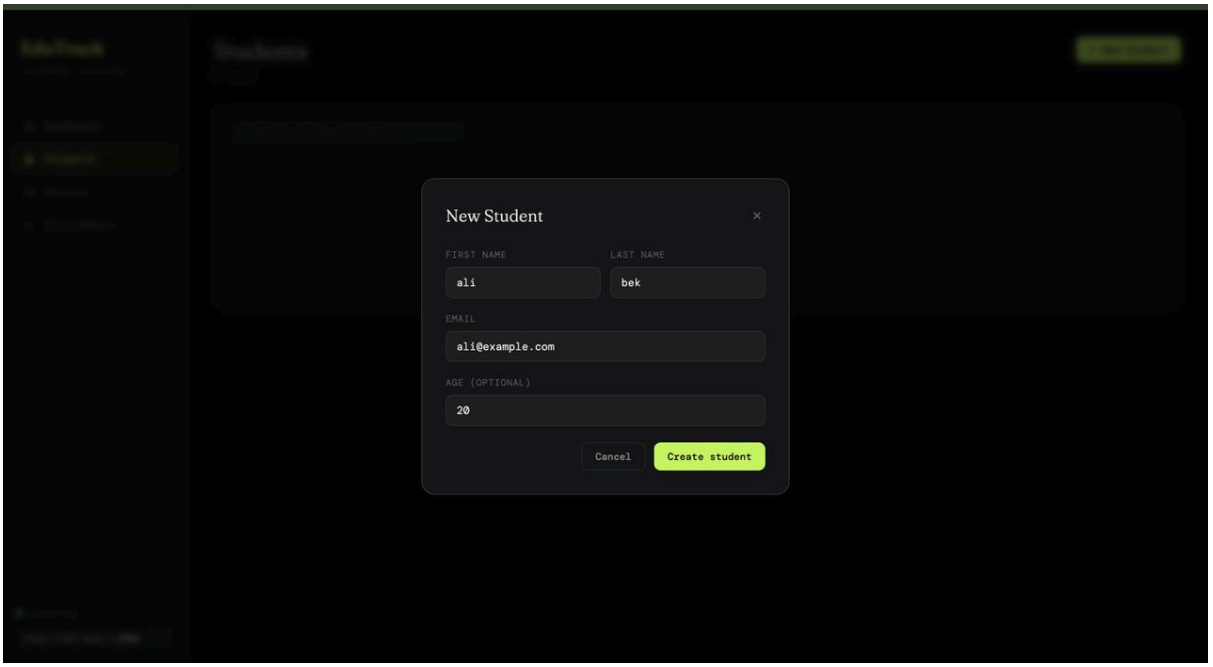
8. Features
Manage students and courses
Search data
Enroll students
Full CRUD operations

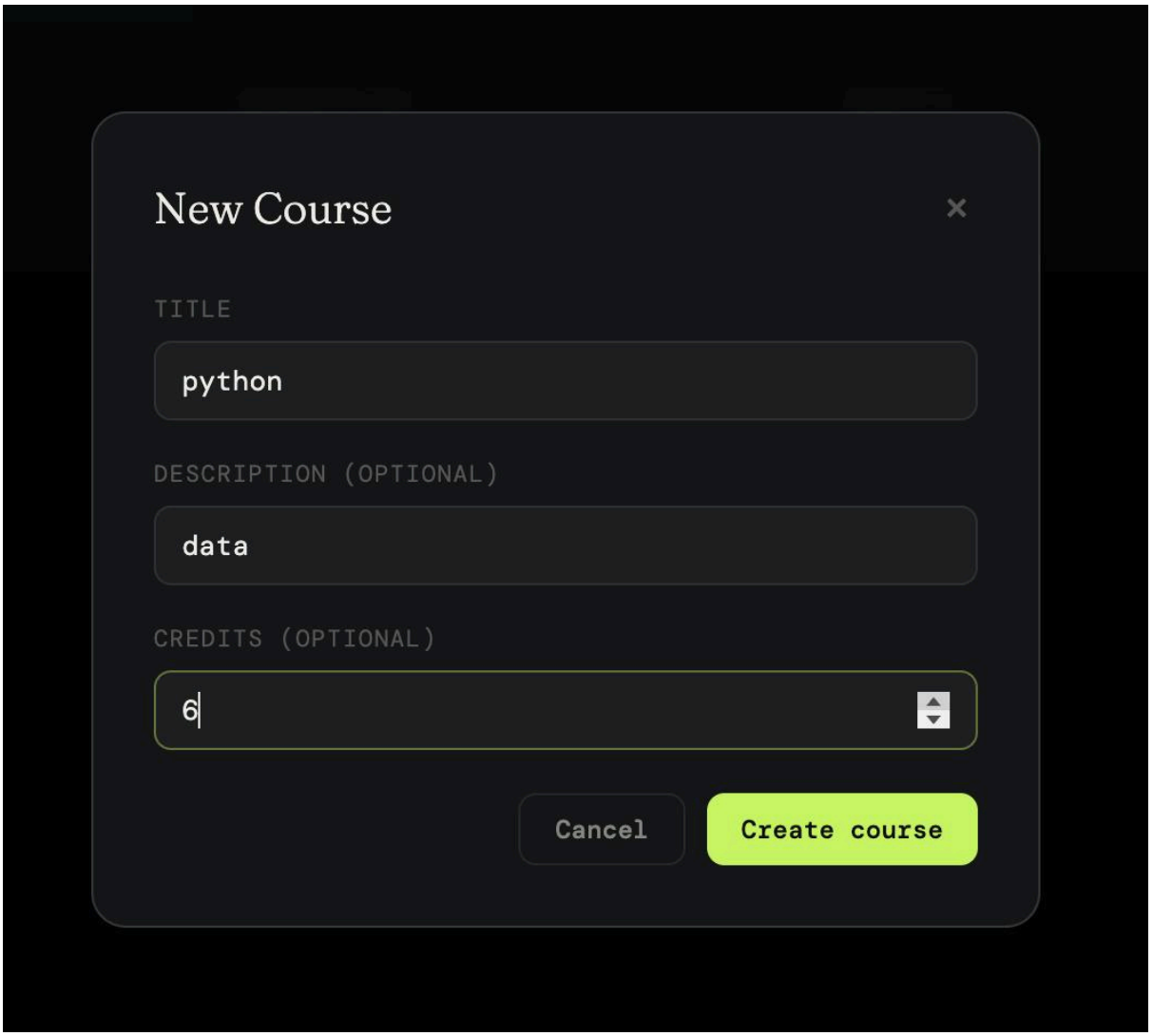
9. HTTP Status Codes
200 OK
201 Created
204 Deleted
400 Bad Request
404 Not Found
422 Validation Error
500 Server Error

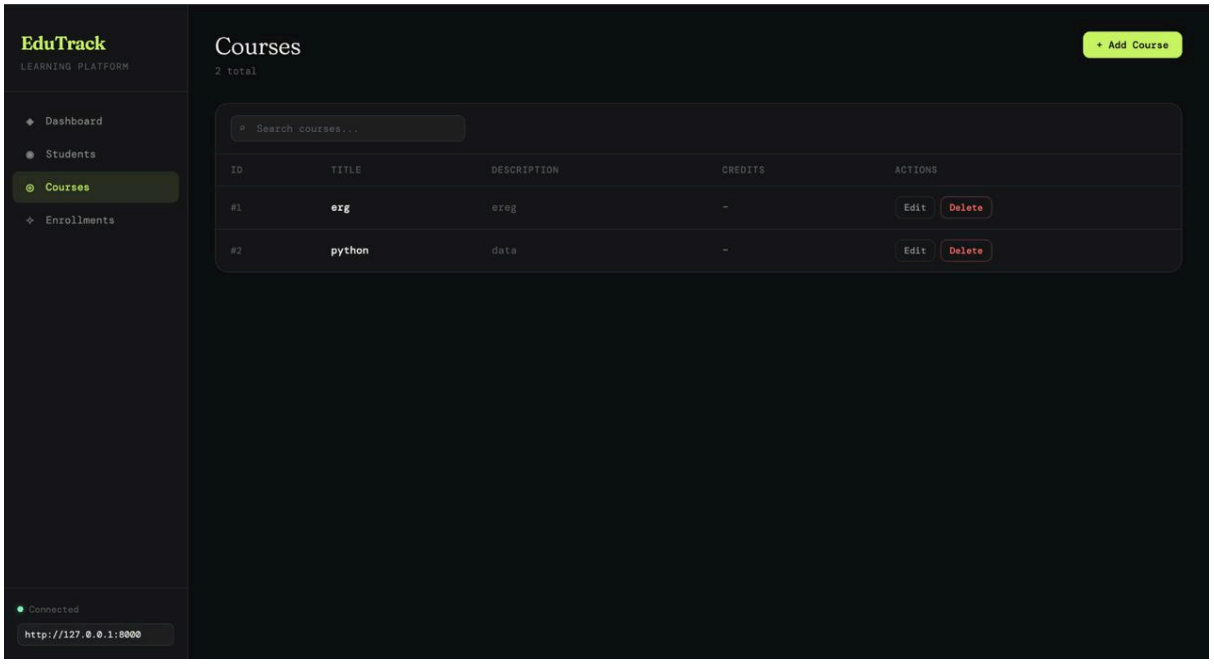
10. CORS
allow_origins=["*"]

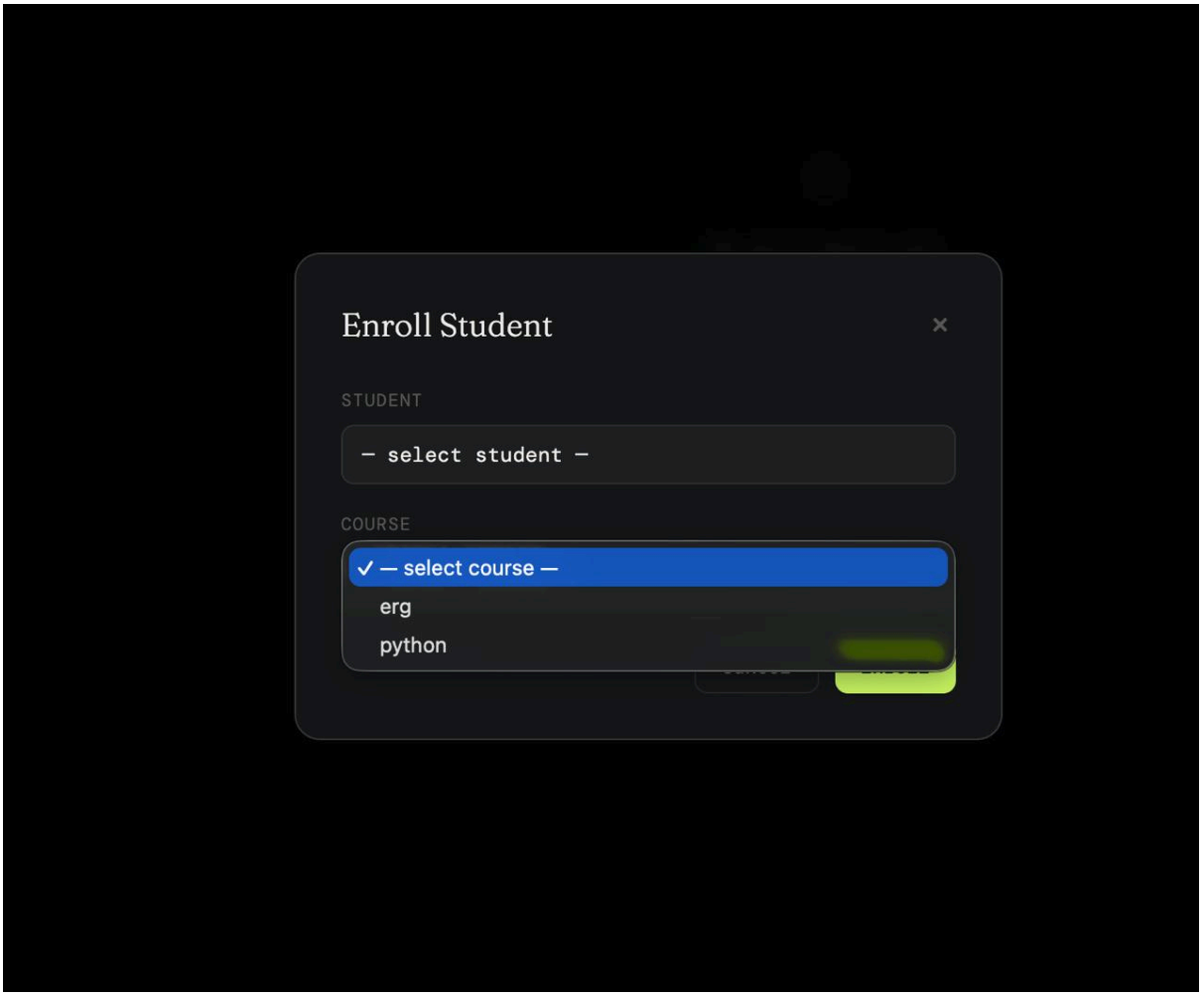

EduTrack
LEARNING PLATFORM
Dashboard
Students
Courses
Enrollments
Connected
http://127.0.0.1:8000
Overview
Updated 00:25:46
↻ Refresh
STUDENTS
0
COURSES
1
ENROLLMENTS
0
RECENT STUDENTS
View all →
No students yet

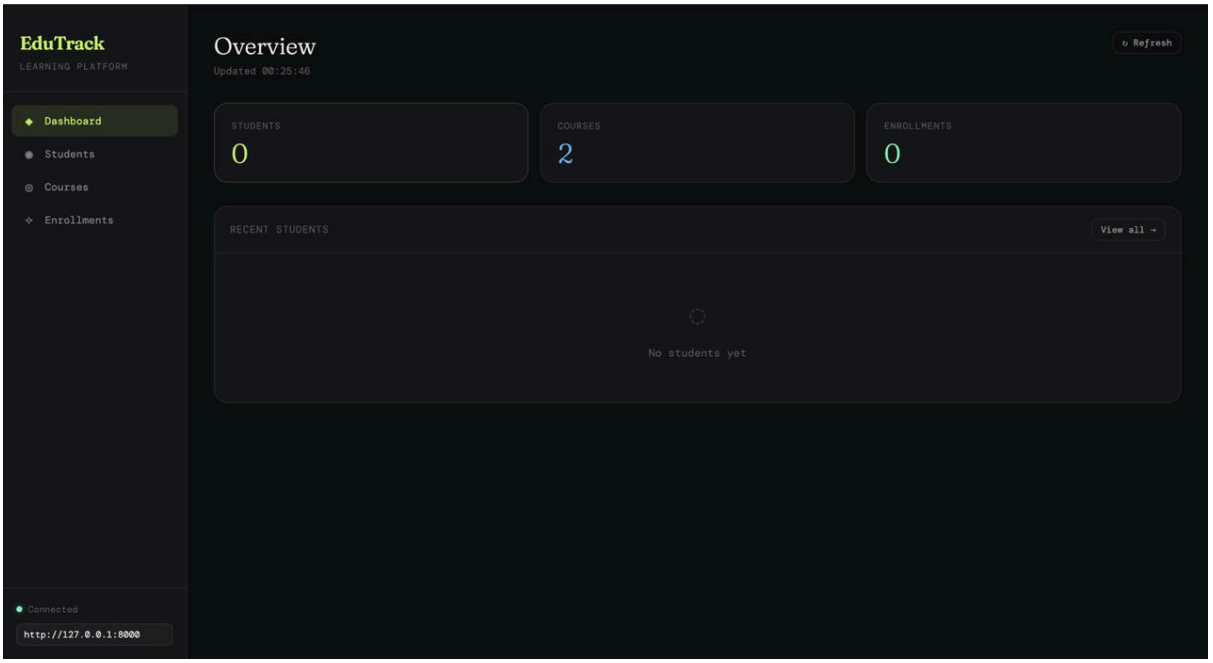

✗ 422: {"detail":
[{"type":"missing","loc":
["body","name"],"msg":"Field
required","input":
{"first_name":"ali","last_name":"bek","email":


New Student
FIRST NAME
ali
LAST NAME
bek
EMAIL
ali@example.com
AGE (OPTIONAL)
20
Cancel
Create student


New Course
TITLE
python
DESCRIPTION (OPTIONAL)
data
CREDITS (OPTIONAL)
6
Cancel
Create course


EduTrack
LEARNING PLATFORM
Dashboard
Students
Courses
Enrollments
Connected
http://127.0.0.1:8000
Courses
2 total
+ Add Course
Search courses...
ID
TITLE
DESCRIPTION
CREDITS
ACTIONS
#1
erg
ereg
-
Edit
Delete
#2
python
data
-
Edit
Delete


Enroll Student
STUDENT
— select student —
COURSE
— select course —
erg
python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